Toronto Metropolitan University

Department of Electrical, Computer, & Biomedical Engineering
Faculty of Engineering & Architectural Science

Course Title:	Electronic Circuits I
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Semester/Year	W2024

Instructor	Dr. Fei Yuan

Assignment/Lab Number:	–
Assignment/Lab Title:	*Design Project*

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Table of Contents

Objectives

- Power supply: $+\mathbf{10}V$ relative to the ground;
- Quiescent current drawn from the power supply: no larger than $\mathbf{10}\ mA$;
- No-load voltage gain (at 1 kHz): $|Avo|$ = **50** (± **10**%);
- Maximum no-load output voltage swing (at 1 kHz): no smaller than 8 V peak to peak;
- Loaded voltage gain (at 1 kHz and with RL = 1 $k\Omega$): no smaller than **90**% of the no-load voltage gain;
- Maximum loaded output voltage swing (at 1 kHz and RL = 1 $k\Omega$): no smaller than 4 V peak to peak;
- Input resistance (at 1 kHz): no smaller than $\mathbf{20}\ k\Omega$;
- Amplifier type: inverting or non-inverting;
- Frequency response: 20 Hz to 50 kHz ($-3dB$ response);
- Type of transistors: BJT;
- Number of transistors (stages): no more than 3;
- Resistances permitted: values smaller than $\mathbf{220}\ k\Omega$ from the E24 series;
- Capacitors permitted: $\mathbf{0.1}\ \mu F$, $\mathbf{1.0}\ \mu F$, $\mathbf{2.2}\ \mu F$, $\mathbf{4.7}\ \mu F$, $\mathbf{10}\ \mu F$, $\mathbf{47}\ \mu F$, $\mathbf{100}\ \mu F$, $\mathbf{220}\ \mu F$;
- Other components (BJTs, diodes, Zener diodes, etc.): only from your ELE404 lab kit.

Circuit Design

Description

The chosen circuit design makes use of three BJTs. The three stages of the designed amplifier are Common-Collector, Common-Emitter, and Common-Collector, respectively. The CE stage is used to attain the desired no-load voltage gain of 50 V/V; the other stage(s) were used for fluctuation prevention and voltage gain regulation/maintenance.

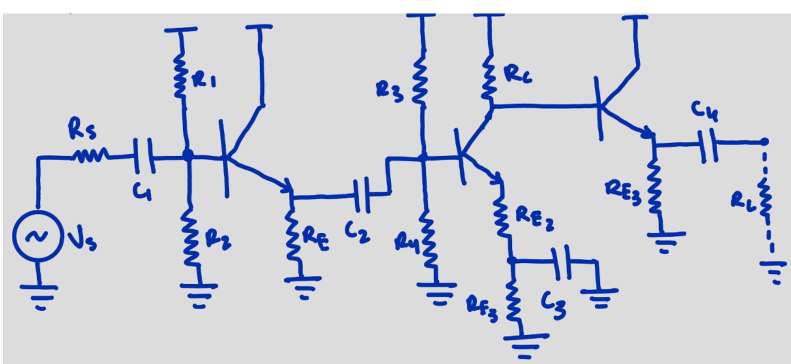


Figure 1: Circuit Design Sketch

Calculations

Stage 1: CC Amplifier

let $I_c = 0.1mA$

$A_v = \frac{V_o}{V_{in}}$ $\quad V_{in} = 0.15$

$1 = \frac{V_o}{0.15}$ (since we want gain to be 1)

$V_o = 0.15V$

$V_o = V_E = R_E I_E$

$R_E = \frac{0.15V}{0.1mA} = 1.5k\Omega$

$V_{BE} = V_B - V_E = 0.7$

$V_B - 0.15 = 0.7$

$V_B = 0.85V$

$V_B = \frac{R_2}{R_1+R_2}(V_{cc})$ [Voltage Division]

$0.85 = \frac{R_2}{R_1+R_2}(10)$

$0.085 = \frac{R_2}{R_1+R_2}$

$R_1 = 0.915R_2$

$R_1 \approx R_2$

Let R_1 & R_2 be $45k\Omega$

$R_{in} = R_1 \| R_2 \| r_{be} + R_E(1+\beta)$

$g_m = \frac{I_c}{V_t} = \frac{0.001}{0.026} = 0.0385$

$R_{in} = \frac{1}{\frac{1}{45K} + \frac{1}{45K} + \frac{1}{(100/0.0385)+(1.5k)(151)}}$ $\quad (\beta = 150)$

$= 20487.86 \geq 20K$

Stage 2: CE Amplifier

Let $I_c = 2mA$

$V_c = \frac{V_{cc} - V_{sat}}{2}$ $\quad \because V_E = \frac{1}{10}V_{cc} = 1V$

$V_{sat} \approx 0.3V$

$V_c = \frac{1}{2}(V_{cc} + V_E)$

$= \frac{1}{2}(10+1)$

$= 5.5$

$5.5 = 10 - R_c(2mA)$

$R_c(2mA) = 10 - 5.5$

$R_c = \frac{4.5V}{2mA}$

$R_c = 2.25k \approx 2.4k\Omega$

The desired gain of 50 will be attained in this stage since the CC stage gain was 1.

$50 = \frac{-g_m R_c}{1 + g_m R_{E_2}}$

$50 + 50 g_m R_{E_2} = -g_m R_c$

$R_{E_2} = \frac{-g_m(2.4k) - 50}{50\, g_m}$

$= 34\Omega \approx 24\Omega$

$V_E = I_E(R_{E_2} + R_{E_3})$

$R_{E_3} = \frac{V_E}{I_E} - R_{E_2}$

$= \frac{1V}{2mA} - 24$

$= 476\Omega \approx 510\Omega$

$V_B = V_{cc}\left(\frac{R_3}{R_3+R_4}\right)$ [Voltage Division]

$V_B = 0.7 + 1$ $\quad V_{cc} = 10$

$0.17 = \frac{R_3}{R_3+R_4}$

$0.17R_3 + 0.17R_4 = R_3$

$0.17R_4 = R_3 - 0.17R_3$

$R_3 \approx 4R_4$

$R_3 = 200K$ $\quad R_4 = 50K$

Stage 3: CC Amplifier (2)

Because of the loading effect we would need $R_{E_4} \approx R_L$ ($1k\Omega$)

let $R_{E_4} = 1.5k\Omega$

$V_B = 8V = V_o = V_E$

$I_E = \frac{V_E}{R_E} = \frac{8}{1500} = 5.33mA \approx I_c$

$I_{c_1} + I_{c_2} + I_{c_3}$ = Total Quiscent current

$(0.1 + 2 + 5.33) = TQC$

$7.43mA = TQC \leq 10mA$

(REQUIREMENT MET) ✓

Since the gain would fluctuate/change when load resistor added, a third (CC) stage was included to eliminate/reduce this effect.

Let $C_2 = C_3 = 220\mu F$

$C_4 = 100\mu F$

$C_1 = 10\mu F$

C_2, C_3 supposed, to maintain gain of CE.

C_1 supposed, due to high input resistance.

C_4 supposed, to block DC current, & provide low resistance path for output of the amplifier.

Simulation

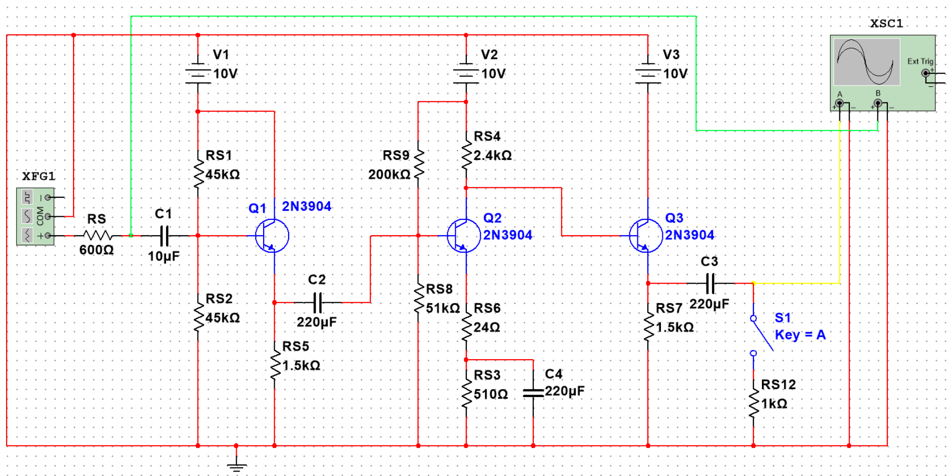


Figure 2: Circuit Design Model (MultiSim)

Stage 1: CC Amplifier

A CC stage was placed in the amplifier first to avoid/minimize signal losses from impedance mismatches and maintain a high input resistance. This is very important when the source can't drive low-impedance loads. The CC stage acts as a buffer, allowing the following stages to amplify a strong, undistorted signal, as will be observed in the second stage.

Stage 2: CE Amplifier

Due to its high voltage gain capacity, a CE amplifier was placed at stage 2. The low input resistance issue of the CE amplifier is avoided due to the preceding CC amplifier isolating the impedance from the source. Without degrading the source signal, the CE stage of this amplifier makes use of its high voltage gain characteristics to meet the necessary gain requirements of this amplifier.

Stage 3: CC Amplifier

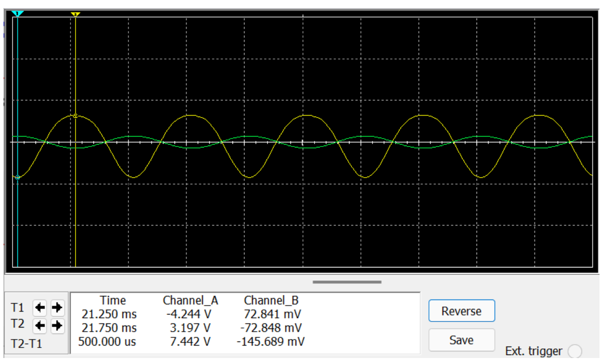
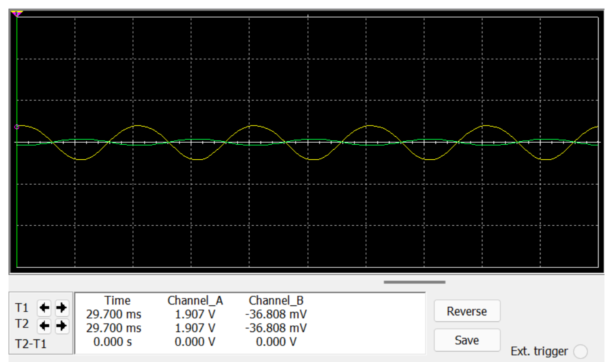
A Common Collector (CC) stage, also known as an emitter follower, was added at the end of this amplifier circuit and acts as a buffer due to its high input and low output impedance. Its buffering effect maintains the overall voltage gain of the amplifier when a load (in this case, a resistor) is connected, as it minimizes the load's impact on the amplification processes of the previous stages. It essentially makes sure a stable amplifier performance is maintained, and the loaded voltage gain is fairly consistent with the no-load one.

Experimental Results

Table 1: Tests for required specifications

Voltage Gain [V/V]	Vs (Input Signal) [mVpp]	Vo (Output Signal) [Vpp]	Voltage [V/V]
Avo	145.575	7.416	50.94
Av	84.223	4.301	51.19

Table 2: Waveforms

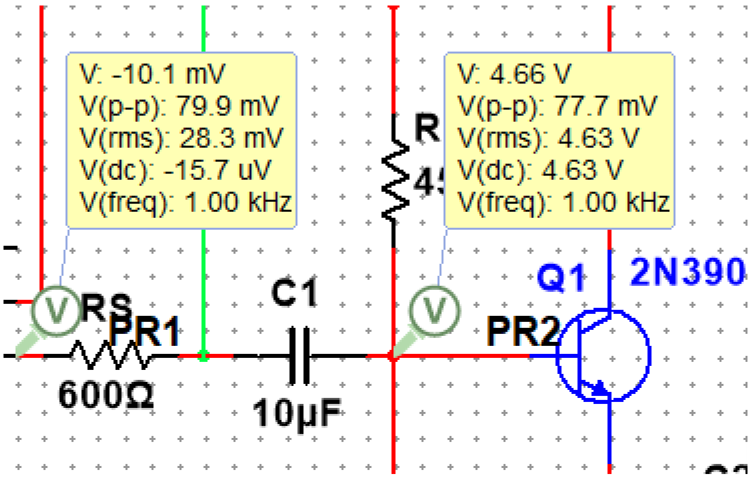
Waveforms	
Without Load	With Load
	

Tables 1 and 2 clearly visualize the fact that experimental results fall within the amplifier specifications. Due to the consistency with the loaded and no-load voltage gain, the amplifier seemed to perform better than expected, making it more suitable for real-world applications.

The no-load output voltage swing, though not exactly above 8 V, is close enough to 8 V, as specified. The loaded output voltage swing, however, was exactly 4.084V, which meets the specifications of the amplifier. At the desired frequency of 1kHz, the voltage gain in both cases also met the requirements, being 90% of each other. The no-load gain was 50.94 and the loaded gain was 51.19, which definitely meets the specifications. Since the output voltage swing was shown to be close to 8 V in Table 2, another graph is not provided.

The input resistance ended up being 21190.91 ohms, which also meets the requirement of being over 20K ohms (using the following values):

$$Rin = \frac{VB \cdot Rs}{Vs - Vb}$$



Conclusions and Remarks

The results show that no substantial difference may exist between the manually computed results and those derived from the simulation. DC node voltages, quiescent current, input resistance, and gain of both circuits were consistent with actual simulations and the calculated values. The difference is only a small one, though that was to be expected since, in most cases, the theoretical computations are conducted by means of idealizing models and simplifying assumptions for complex real phenomena, hence very far from reality when measured in laboratory or field conditions. Besides, even when in the same order of magnitude, the resistance value tolerance used for input and output resistance may vary. So, all these further jeopardize the precision of these computations. The aim of this project was to design a multistage amplifier using BJT transistors. All the specifications pertaining to the design of the amplifier were realized successfully, whereby the results of manual computations and from tests in simulation both agreed. In general, the project was successful.